

Does Synthetic Bulk RNA-seq Data Protect Donors? Privacy Auditing through Membership Inference Attacks

Charlene Jarrell, Daniil Filienko, Jonathan Kim, Emma Szebenyi,
Sikha Pentyala, Steven Golob, Martine De Cock

University of Washington Tacoma

1 Introduction

Synthetic data generation (SDG) [6, 11] has emerged as a promising approach for enabling data sharing in sensitive domains such as biomedical research [9, 10]. By generating artificial data that preserve the statistical and biological characteristics of real datasets, SDG can support downstream analyses, data augmentation, and the development of more robust and fair models while reducing direct exposure of individual-level data [7]. However, synthetic data are not inherently privacy-preserving, especially when generative models are optimized for high fidelity, which may inadvertently encode information about training samples [14]. Synthetic datasets must therefore be evaluated for privacy risks alongside fidelity and utility before release.

Membership Inference Attacks (MIAs) have emerged as a standard approach for empirically auditing the privacy of AI models [12]. In a MIA, an adversary attempts to determine whether a specific individual’s data was included in the training set of a model. This is critical because, if a donor’s data can be linked to the training of a synthetic cancer dataset, for example, then this would reveal that the person has cancer [16]. While MIAs may not directly reveal sensitive attributes, they serve as a tractable lower bound on overall privacy risk and a proxy for more severe attacks such as attribute inference and data reconstruction. Despite extensive study of MIAs in image models [12], language models [1], and synthetic tabular data [5, 14, 15], comparatively little work has explored MIAs in transcriptomics such as synthetic bulk RNA-seq data. The Critical Assessment of Massive Data Analysis (CAMDA) Health Privacy Challenge at ISMB2025 highlighted this gap explicitly, calling for privacy audits of synthetic bulk RNA-seq data, and none of its participants succeeded in mounting an effective adversarial attack. In a subsequent overview paper on the CAMDA 2025 challenge [10], a SDG-method-agnostic GAN-Leaks attack [2] achieved elevated attack performance against CVAE-generated data ($AUC \approx 0.7$), moderate performance on multivariate-based-generated data ($AUC \approx 0.53$), and near-random performance on differentially private (DP) and lower-utility models.

In this paper, we demonstrate that it is possible to achieve substantially higher attack success rates when leveraging knowledge about the SDG method that produced the synthetic data. We propose strong MIAs for synthetic bulk RNA-seq data produced by generators from the red teaming track of the CAMDA 2026 challenge¹: MELOMIA (Measuring Losses MIA) for deep learning generators, and MAHALAMIA (Mahalanobis distance MIA) and MAMA-MIA [5] for statistical and probabilistic generators respectively. Using the CAMDA evaluation pipeline, we systematically assess privacy leakage across generative models and find that high-fidelity non-DP generators are substantially more susceptible to tailored MIAs (AUC up to 0.922), while DP provides meaningful resistance underscoring that formal privacy guarantees remain essential alongside fidelity for safe synthetic data release.

CAMDA 2026 Track I Red Team – Threat Model. The goal of this challenge was to “launch membership inference attacks (MIA) against selected blue teams’ solutions from the Health Privacy Challenge 2025”. The challenge dataset consists of synthetic data derived for two pre-processed

¹<https://benchmarks.elsa-ai.eu/?ch=8>

TCGA bulk RNA-seq cohorts: TCGA-BRCA (1,089 samples, 978 genes, 5 cancer subtypes) and TCGA-COMBINED (4,323 samples, 978 genes, 10 tissue origins). Each real TCGA dataset D_{real} was split 80%-20% into D_{train} and D_{test} . Each D_{train} was subsequently used to generate a synthetic dataset D_{synth} using four blue-team generators from the 2025 challenge: a CVAE baseline [13], a Multivariate generator (MVN) baseline [10], an embedded noisy diffusion (ND) model based on [4], and a differentially private probabilistic graphical model (DP-PGM, $\varepsilon = 10$) [3, 8], amounting to eight synthetic datasets in total (four for each of TCGA-BRCA and TCGA-COMBINED).

The platform evaluates under a black-box model where an adversary has the released synthetic data D_{synth} and has access to public implementations of each generator’s architecture and training procedure, but cannot access the trained model parameters or the original training data D_{train} . An additional auxiliary reference set D_{aux} of known non-members is provided only for TCGA-COMBINED. The adversary must infer for each real sample x in D_{real} whether x was included in D_{train} . In the following sections, we describe our attack methodology and summarize the resulting privacy findings.

2 Methodology

2.1 Attacks against deep learning generators

We introduce MELOMIA (Measurement of Losses based MIA) that adapts the loss-trajectory framework of [15] to synthetic bulk RNA-seq data. The key idea is to extract per-sample loss features from shadow models across different noise levels and timesteps, and train a meta-classifier $\mathcal{M}$ to predict membership from these loss patterns. A central requirement is that the shadow models used to train $\mathcal{M}$ must themselves be trained on *synthetic* data. At inference time, the adversary only has access to the released synthetic dataset D_{synth} , from which a proxy model can be trained; consequently, the loss distributions observed during training and inference must be aligned. To achieve this, we introduce **synth-shadow modeling**. We first train *base shadow* generators on K random 80–20 splits of D_{real} and use them to generate K internal synthetic datasets. We then train a second layer of shadow models, the *synth-shadows*, exclusively on these synthetic datasets for feature extraction. Thus, real data enters the pipeline only through the base shadows, while both the synth-shadows and the inference-time proxy operate entirely in the synthetic domain.² Concretely, the attack proceeds in five steps:

1. **Generate internal synthetic datasets.** We sample K random 80–20 splits of D_{real} . For each $k = \{1, \dots, K\}$, we train a base shadow generator on D_{tr}^k and use it to produce an internal synthetic dataset D_{syn}^k .
2. **Train synth-shadow models.** We train K synth-shadow models, one on each D_{syn}^k . The membership label of a real sample $x \in D_{real}$ with respect to synth-shadow k is inherited from the corresponding base shadow: x is a member iff $x \in D_{tr}^k$.
3. **Extract loss features.** For each candidate sample $x \in D_{real}$ and each synth-shadow k , we compute a per-sample loss feature $\phi_k(x)$ (the form of ϕ is generator-specific; see below) and pair it with x ’s inherited membership label. Note that since each sample is passed through all K synth-shadow models, it is associated with K potentially different membership labels.
4. **Train the meta-classifier.** Pooling all $K \cdot |D_{real}|$ feature–label pairs, we train a meta-classifier $\mathcal{M}$ that maps loss features to a membership prediction.
5. **Inference.** Given D_{synth} and a target sample $x \in D_{real}$, we train a final proxy generator on D_{synth} — substituting for the unobserved target model — compute $\phi(x)$ using this proxy, and apply $\mathcal{M}$ to predict whether $x \in D_{train}$.

²Base shadows use the same hyperparameters as the target; synth-shadows and the proxy can use enhanced settings to fit as tightly as possible to the synthetic data.

The two generators differ only in the form of $\phi(x)$ computed in steps 3 and 5.

NoisyDiffusion (ND). The loss feature is a $T \times N$ matrix of mean-squared errors (MSEs) between the synth-shadow’s predicted noise and the injected noise, with $T = 15$ timesteps and $N = 600$ noise vectors ε drawn once with a fixed seed and reused across all samples and models. For each (t, ε) pair, we form $x_t = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \varepsilon$, query the noise-prediction head for $\hat{\varepsilon}$, and record $\|\hat{\varepsilon} - \varepsilon\|^2$ averaged over 978 gene dimensions. $\mathcal{M}$ is an ensemble (XGBoost, RF, LightGBM, CatBoost, MLP) blended by softmax-weighted AUC. We use Bayesian optimisation (Optuna) to tune each classifier’s hyperparameters and timestep subset $\mathcal{T} \subseteq \{1, \dots, T\}$, collapsing the $|\mathcal{T}| \times N$ slice into per-timestep statistics and gradients to yield an optimised summary feature per classifier. The final membership score is a gated weighted ensemble of these specialised classifiers.

Conditional VAE (CVAE). The CVAE has no diffusion timestep axis, so we replace the timestep sweep with a temperature sweep over the encoded posterior. For each candidate sample x , we obtain encoder parameters (μ, σ) and, for each temperature $\alpha \in \{0, 0.5, 1.0, 1.5, 2.0, 3.0\}$ and each of $N = 50$ frozen noise vectors ε , draw $z = \mu + \alpha\sigma\varepsilon$, decode to $\hat{x}$, and record the per-sample reconstruction MSE. We additionally concatenate the 128-dimensional per-latent KL-divergence contribution between the encoded posterior and the standard-normal prior, since training members typically exhibit sharper posteriors and therefore more distinctive per-dimension KL patterns. This combined vector constitutes $\phi_k(x)$ for CVAE and is summarised analogously to ND before being passed to $\mathcal{M}$.

2.2 Attacks against statistical and probabilistic generators

Statistical generators encode their learned distributional structure in explicit parameters such as covariances for MVN, marginals for DP-PGM. Both attacks in this category follow MAMA-MIA [5] and exploit this structure by computing a domain-ratio membership score that measures how well a candidate sample’s features are explained by the synthetic model relative to a reference distribution.

Multivariate Generator (MVN). The MVN generator models D_{synth} as a multivariate Gaussian over the 978 gene dimensions, making the covariance structure of D_{synth} the natural membership signal. We introduce MAHALAMIA (Mahalanobis distance based MIA), which scores each candidate sample $x \in D_{\text{real}}$ by its Mahalanobis distance to the synthetic distribution, $d_{\text{syn}}(x) = \sqrt{(x - \mu_{\text{syn}})^\top \Sigma_{\text{syn}}^{-1} (x - \mu_{\text{syn}})}$ where μ_{syn} and Σ_{syn} are the empirical mean and covariance of D_{synth} . Samples with lower $d_{\text{syn}}(x)$ receive higher membership scores. When the auxiliary reference set D_{aux} is available (TCGA-COMBINED), we additionally compute $d_{\text{aux}}(x)$ against the distribution of D_{aux} and use the normalised ratio $d_{\text{aux}}(x)/d_{\text{syn}}(x)$ as the final membership score, which is high when x lies close to the synthetic distribution relative to the auxiliary distribution.

Private PGM (DP-PGM). DP-PGM explicitly learns low-order marginals of D_{train} under DP, making these marginals the natural attack surface. We therefore apply MAMA-MIA [5], which scores a candidate sample x using domain ratios computed from synthetic marginals relative to a population baseline. We target both one-way marginals (a) per-gene distributions $\hat{p}(g_i)$ for $i = \{1, \dots, 978\}$ and subtype marginal $\hat{p}(s)$; (b) two-way gene–subtype marginals $\hat{p}(g_i, s)$. For a sample x with subtype label s_x , MAMA-MIA aggregates these domain ratios across all targeted marginals to produce a final membership score, assigning higher scores to samples whose gene–subtype patterns are over-represented in the synthetic distribution. Since subtype labels are unavailable in the provided reference set, we use the full labeled data pool as auxiliary data for estimating the required marginals.

Table 1: MIA evaluation on TCGA-BRCA and TCGA-COMBINED. AUC-ROC (**AUC**), precision-recall AUC (**AUPR**), TPR at FPR=1% (**T@1**) and 10% (**T@10**). Higher is better; ≈ 0.5 AUC indicates near-random. Scores averaged over 5 splits. **Ours**: MAHALAMIA (MVN), MeLoMIA (CVAE, ND), MAMA-MIA (DP-PGM).

Dataset	Method	MVN (noise = 0.7)				CVAE				NoisyDiffusion				DP-PGM ($\varepsilon = 10$)			
		AUC	AUPR	T@1	T@10	AUC	AUPR	T@1	T@10	AUC	AUPR	T@1	T@10	AUC	AUPR	T@1	T@10
BRCA	MC	0.508	0.816	0.027	0.818	0.562	0.912	1.000	1.000	0.515	0.806	0.020	1.000	0.500	0.801	0.016	0.449
	GAN-leaks	0.533	0.818	0.019	0.121	0.706	0.916	0.277	0.427	0.534	0.824	0.032	0.137	0.505	0.803	0.012	0.098
	Conf-LR	0.568	0.818	0.005	0.093	0.549	0.816	0.013	0.111	0.539	0.809	0.008	0.097	0.508	0.805	0.013	0.106
	Conf-RF	0.512	0.807	0.017	0.112	0.538	0.820	0.019	0.123	0.512	0.808	0.012	0.101	0.511	0.808	0.023	0.116
	Ours	0.922	0.981	0.698	0.825	0.753	0.928	0.274	0.489	0.620	0.858	0.045	0.193	0.528	0.824	0.036	0.142
COMBINED	MC	0.505	0.809	0.012	0.820	0.525	0.843	0.016	1.000	0.508	0.803	0.013	1.000	0.502	0.803	0.011	0.639
	LOGAN [†]	0.502	0.802	0.0107	0.103	0.502	0.801	0.011	0.105	0.502	0.801	0.009	0.101	0.500	0.800	0.012	0.102
	GAN-leaks	0.528	0.811	0.012	0.110	0.635	0.860	0.021	0.186	0.514	0.809	0.014	0.115	0.503	0.802	0.012	0.104
	GAN-leaks-cal [†]	0.566	0.847	0.164	0.172	0.560	0.902	0.908	0.908	0.522	0.808	0.679	0.679	0.505	0.804	0.012	0.107
	DOMIAS-KDE [†]	0.500	0.800	0.009	0.100	0.500	0.800	0.011	0.100	0.499	0.799	0.011	0.099	0.500	0.801	0.014	0.100
	Conf-LR	0.514	0.804	0.010	0.105	0.508	0.802	0.008	0.100	0.510	0.806	0.012	0.109	0.504	0.802	0.012	0.103
	Conf-RF	0.512	0.807	0.011	0.114	0.512	0.806	0.013	0.107	0.507	0.805	0.012	0.106	0.507	0.804	0.010	0.103
	Ours	0.899	0.972	0.413	0.729	0.677	0.872	0.022	0.194	0.7383	0.7426	0.1006	0.3823	0.556	0.840	0.038	0.178

3 Results

We evaluate by splitting D_{real} into 5 random 80–20 partitions and training all four generators independently on each training split, yielding 5 synthetic datasets per generator; this ensures availability of ground-truth labels for internal evaluation. Each synthetic data is attacked using the methods in Section 2 and metrics are averaged across partitions.

Table 1 reports results for our SDG-aware attacks alongside SDG-method-agnostic baselines from the CAMDA 2026 challenge. Our attacks substantially outperform all baselines on every generator, confirming that knowledge of the generative mechanism is critical for effective privacy auditing. Vulnerability closely tracks the fidelity–privacy trade-off visible in Figure 1. Non-DP generators (MVN, CVAE, ND), which produce synthetic data with broad distributional overlap with D_{real} , are all susceptible, whereas DP-PGM, whose tighter synthetic clusters reflect the fidelity cost of DP, provides substantially stronger protection. Nevertheless, MAMA-MIA still achieves a consistent improvement above all baselines on both cohorts, suggesting that formal DP at $\varepsilon = 10$ leaves residual vulnerability. Results are overall consistent across both datasets but varies by generator across datasets: MVN and CVAE attacks are modestly weaker on the larger TCGA-COMBINED, while ND and DP-PGM attacks perform comparably or better on COMBINED.

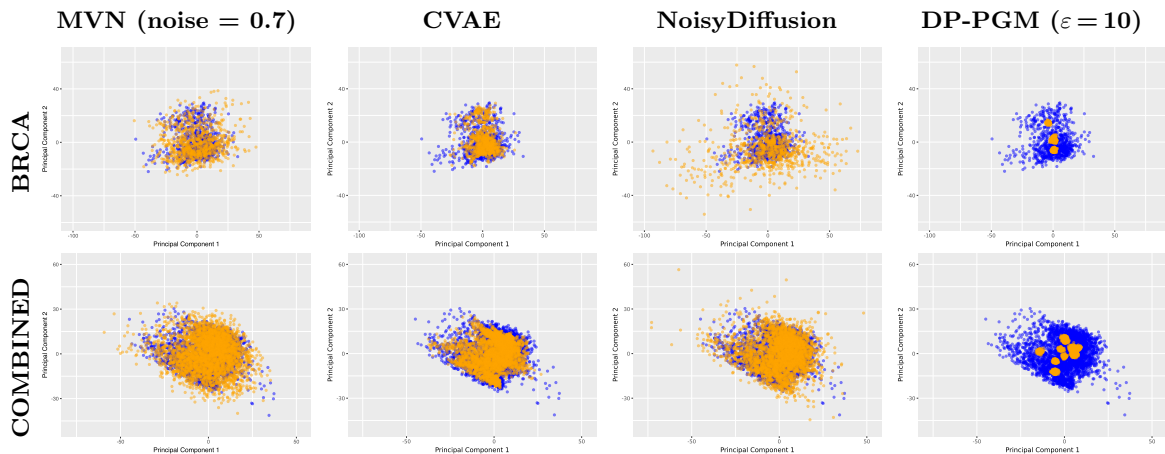


Figure 1: PCA projections of real (●) and synthetic (●) data for split 1 (seed 42) of the internal evaluation set. Good overlap between real and synthetic points indicates high fidelity.

We additionally evaluate MAHALAMIA as a model-agnostic attack by applying it without modification to synthetic data from the 2 other non-DP generators (CVAE and ND). On CVAE-generated data, MAHALAMIA achieves AUC = 0.986 on BRCA — exceeding the model-specific MELoMIA on the same generator (AUC = 0.753) — and AUC = 0.583 on COMBINED, suggesting that CVAE implicitly preserves the covariance structure of its training data, particularly on smaller cohorts. Performance on ND is more variable (BRCA: AUC = 0.489; COMBINED: AUC = 0.753), indicating that diffusion-based generation preserves covariance less consistently, likely due to the stochastic noise injection process.

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